

Olympiad Test Paper — Mathematics (Class 7)

Chapter 13: Connecting the Dots

Subject	Mathematics (NCERT Class 7)
Chapter	13 — Connecting the Dots (Data Handling)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember that mean = total ÷ count, the median needs the data SORTED first, and the mode is the most frequent value — they are three different summaries, not the same "average." Watch the classic traps: comparing totals when group sizes differ, forgetting to sort before taking a median, mistaking the mode for the middle, and reading a bar graph's shape without checking its scale (especially a truncated axis). Do not write on this paper — mark answers on your answer sheet.

1. Which of these statements is a "statistical" statement — a likely conclusion drawn from data, not a certainty?

- (A) A person 6 ft tall is more likely to be an adult than a toddler.
- (B) A triangle has exactly three sides.
- (C) The sum of 12 and 8 is 20.
- (D) A square has four equal sides.

2. Shubman scored 23, 7, 10, 52, 18 in 5 matches (total 110); Yashasvi scored 26, 53, 2, 15 in 4 matches (total 96). Why is comparing only the totals unfair here?

- (A) Totals are never useful in cricket.
- (B) Shubman's highest score is the only thing that matters.
- (C) Yashasvi's scores are all even.
- (D) They played different numbers of matches, so more matches allow a larger total — compare the per-match mean instead.

3. The mean of 0, 17, 21, 90 is:

- (A) 19
- (B) 90
- (C) 32
- (D) 128

4. The median of 0, 17, 21, 90 is:

- (A) 19 (B) 21
(C) 17 (D) 32

5. The mode of 5, 3, 5, 7, 5, 8, 3 is:

- (A) 3 (B) 7
(C) 5 (D) 8

6. The range of the marks 45, 78, 33, 90, 60 is:

- (A) 90 (B) 45
(C) 33 (D) 57

7. The mean of five numbers is 20. Four of them are 18, 22, 15 and 25. The fifth number is:

- (A) 20 (B) 80
(C) 18 (D) 25

8. A shopkeeper wants to decide which shoe size to stock the most of. Which representative value should he use?

- (A) Mean (B) Median
(C) Mode (D) Range

9. To find the median of 7, 3, 9, 5, 11, 4 a student must FIRST:

- (A) add all the values (B) sort the values in order
(C) find the most frequent value (D) divide by 6

10. The median of 7, 3, 9, 5, 11, 4 is:

- (A) 5 (B) 5.5
(C) 6 (D) 7

11. A bar graph uses the scale 1 cm = 10 runs. A player's bar is 4.5 cm tall. The runs scored are:

- (A) 45 (B) 14.5
(C) 4.5 (D) 450

12. Using the scale 1 cm = 10 runs, a player who scored 75 runs needs a bar of height:

- (A) 75 cm (B) 7.5 cm
(C) 0.75 cm (D) 750 cm

13. The mean of the first five whole numbers 0, 1, 2, 3, 4 is:

- (A) 2.5 (B) 10
(C) 3 (D) 2

14. The mean of 12, 15, 18, 11, 14 is:

- (A) 14 (B) 15
(C) 12 (D) 70

15. The mode of 4, 7, 4, 9, 7, 4, 2 is:

- (A) 7 (B) 9
(C) 4 (D) 2

16. Player A scored 28, 31, 29, 32 and Player B scored 5, 55, 2, 58. Both have a mean of 30. Which is more consistent, and why?

- (A) Player A, because the range (4) is much smaller than B's range (56).
(B) Player B, because the total is the same.
(C) Player B, because B has the highest single score.
(D) They are equally consistent because the means are equal.

17. The seven daily temperatures ($^{\circ}\text{C}$) were 30, 32, 31, 30, 33, 30, 35. The mode is:

- (A) 31 (B) 30
(C) 32 (D) 35

18. For the temperatures 30, 32, 31, 30, 33, 30, 35 ($^{\circ}\text{C}$), the median is:

- (A) 30 (B) 32
(C) 33 (D) 31

19. Five test scores are 60, 60, 70, 80, 80. Which statement is TRUE?

- (A) The data has a single mode. (B) The mode is 70.
(C) The mean is 70 and the median is 70. (D) The range is 80.

20. The mean weight of 4 boys is 35 kg. If a fifth boy weighing 45 kg joins, the new mean weight is:

- (A) 37 kg (B) 40 kg
(C) 36 kg (D) 45 kg

21. A data set of 6 values has mean 12. If every value is increased by 4, the new mean is:

- (A) 12 (B) 48
(C) 16 (D) 72

22. The mean of 8 numbers is 21. If one number, 14, is removed, the mean of the remaining 7 numbers is:

- (A) 21 (B) 22
(C) 20 (D) 8

23. A bar graph has bars for four fruits: Apple 25, Banana 40, Mango 15, Orange 20 (sales in kg). Which fruit's bar is tallest, and by how much does it beat the shortest?

- (A) Banana; 25 kg taller than the shortest (B) Apple; 10 kg taller than the shortest
(C) Mango; 15 kg taller than the shortest (D) Banana; 40 kg taller than the shortest

24. The median of the four values 12, 7, 9, 20 is:

- (A) 9 (B) 14.5
(C) 12 (D) 10.5

25. Which set of numbers has its mean, median and mode ALL equal to 5?

(A) 3, 4, 5, 7

(B) 5, 5, 5, 5

(C) 1, 5, 9

(D) 4, 5, 6, 8

26. A bar graph's vertical axis starts at 90 instead of 0. A value rises from 92 to 96. Why does the bar look like it nearly doubled?

(A) Because 96 is double 48.

(B) Because $96 - 92 = 4$ is a large jump.

(C) Because the eye reads bar heights as if measured from zero, so a truncated axis exaggerates the difference.

(D) Because bar graphs always exaggerate.

27. In the list 2, 4, 4, 4, 6, 100 which representative value is MOST distorted (pulled away from the typical value) by the 100?

(A) The mode

(B) The median

(C) The mean

(D) The smallest value

28. Seven students' marks are 8, 6, 7, 9, 6, 10, 5. The mean (to one decimal place) is:

(A) 7.3

(B) 7.0

(C) 6.0

(D) 51

29. The mean of six numbers is 15. Five of them are 10, 12, 18, 20 and 14. The sixth number is:

(A) 16

(B) 15

(C) 90

(D) 14

30. A pictograph uses one symbol = 5 books. A shelf shows $6\frac{1}{2}$ symbols. The number of books is:

(A) 11

(B) 30

(C) 65

(D) 32.5

31. The runs of five players read off a bar graph are 30, 45, 45, 60 and 20. The mode of these runs is:

(A) 30

(B) 45

(C) 60

(D) 20

32. Which is the correct order of these representative values for the data 2, 2, 2, 5, 8, 100?

(A) mean < median < mode

(B) mean = median = mode

(C) mode < median < mean

(D) median < mode < mean

33. A double bar graph compares two classes' favourite-sport counts. For Cricket, Class A = 18 and Class B = 12; for Football, Class A = 10 and Class B = 22. Which sport shows the LARGER gap between the two classes?

(A) Cricket, gap 6

(B) Football, gap 32

(C) Cricket, gap 18

(D) Football, gap 12

- 34.** The mean of 10 numbers was found to be 25. Later it was discovered that one number was read as 36 instead of the correct 56. The corrected mean is:
- (A) 25 (B) 23
(C) 27 (D) 45
- 35.** A class of 30 students has a mean height of 140 cm; another class of 20 students has a mean height of 150 cm. The mean height of all 50 students is:
- (A) 145 cm (B) 144 cm
(C) 146 cm (D) 290 cm
- 36.** Six observations are arranged as 4, 7, x, 13, 16, 20 in increasing order, and their median is 11. The value of x is:
- (A) 9 (B) 11
(C) 10 (D) 13
- 37.** The mode of the data 2, 2, 3, 3, 5, 5, 7 is:
- (A) 2 (B) 3
(C) 5
(D) There is no single mode (2, 3 and 5 each appear twice)
- 38.** Reading a bar graph, monthly rainfall (mm) is Jun 80, Jul 120, Aug 100, Sep 60. The mean monthly rainfall is:
- (A) 80 mm (B) 90 mm
(C) 100 mm (D) 360 mm
- 39.** For the data 5, 8, 8, 11, 13 the value of (mean – median) is:
- (A) 1 (B) 0
(C) -1 (D) 8
- 40.** A batter's scores are 0, 17, 21, 90. Another batter's scores are 67, 55, 18, 35. Comparing their ranges, who is the MORE consistent batter?
- (A) The first batter, range 90 (B) The first batter, range 49
(C) They are equally consistent (D) The second batter, range 49
- 41.** The mean of 9 numbers is 30. If three more numbers with mean 42 are added, the mean of all 12 numbers is:
- (A) 36 (B) 33
(C) 35 (D) 72
- 42.** A data set is 10, 14, 16, 16, 19, 21 (sorted). If the value 14 is changed to 27, which measure does NOT change at all?
- (A) The mean (B) The range
(C) The median (D) The mode
- 43.** Eight friends' ages are 11, 12, 12, 13, 13, 13, 14, 16. Which statement is TRUE?

(A) mode = 12, median = 13

(B) mode = 13, median = 13

(C) mode = 13, median = 12.5

(D) mode = 14, median = 13

44. A survey records the number of pets per house: 0, 1, 0, 2, 1, 0, 3, 1, 0. The mode number of pets is:

(A) 0

(B) 1

(C) 2

(D) 3

45. The mean of five consecutive whole numbers is 12. The largest of them is:

(A) 12

(B) 16

(C) 14

(D) 10

46. A bar graph shows weekly savings (₹): Mon 20, Tue 0, Wed 40, Thu 30, Fri 10. The mean DAILY saving over these five days is:

(A) ₹25

(B) ₹20

(C) ₹30

(D) ₹100

47. The median of the seven values 14, 9, 21, 7, 18, 12, 9 is:

(A) 9

(B) 18

(C) 14

(D) 12

48. In a data set the mean is much greater than the median. This most likely indicates:

(A) a few unusually large values (outliers) pulling the mean up

(B) all values are equal

(C) the data was not sorted

(D) the mode equals the mean

49. Five numbers have a mean of 8. Four of them are 6, 9, 7 and 11. The fifth number is:

(A) 8

(B) 7

(C) 40

(D) 5

50. A pictograph of cars sold uses one car symbol = 4 cars. Showroom X shows 7 symbols, Showroom Y shows $5\frac{1}{2}$ symbols. How many more cars did X sell than Y?

(A) 4

(B) 8

(C) 1.5

(D) 6

51. For the data 3, 7, 7, 7, 9, 11, 12 the mode, median and mean are respectively:

(A) 7, 7, 8

(B) 7, 8, 7

(C) 9, 7, 8

(D) 7, 7, 7

52. Two sections both have mean marks 60. Section P has range 8; Section Q has range 40. Which conclusion is correct?

(A) Section Q's marks are more tightly clustered.

(B) Section P's marks are more consistent (less spread out).

(C) Section P must have the higher total.

(D) The means being equal means the marks are identical.

- 53.** The mean of 6, 10, 14, x and 20 is 13. The value of x is:
- (A) 13 (B) 65
(C) 50 (D) 15
- 54.** A bar graph (scale 1 unit = 2 students) shows the bar for "walks to school" reaching 9 units. The number of students who walk to school is:
- (A) 9 (B) 11
(C) 18 (D) 4.5
- 55.** The marks of 9 students are 10, 12, 12, 14, 15, 15, 15, 18, 19. The difference (mode – median) is:
- (A) 0 (B) 2
(C) 1 (D) 3
- 56.** A data set of 5 numbers has mean 14 and range 12, with smallest value 8. The largest value is:
- (A) 14 (B) 20
(C) 26 (D) 12
- 57.** The mean of three numbers is 10 and the mean of two other numbers is 15. The mean of all five numbers together is:
- (A) 12.5 (B) 13
(C) 12 (D) 25
- 58.** A bar graph compares trees planted: 2021 $\rightarrow$ 30, 2022 $\rightarrow$ 45, 2023 $\rightarrow$ 60, 2024 $\rightarrow$ 45. Which value is both the mode and equal to the median of the four numbers?
- (A) 30 (B) 60
(C) 45 (D) 60 is the median, 45 is the mode
- 59.** A list of 6 numbers reads 5, 9, 11, 11, 14, x . The mode is 11 and the mean is 11. The value of x is:
- (A) 11 (B) 16
(C) 14 (D) 9
- 60.** A magazine shows a bar graph where the vertical axis starts at 50, and Brand A's bar (value 54) looks twice as tall as Brand B's bar (value 52). The honest conclusion is:
- (A) Brand A sold about twice as much as Brand B.
(B) The graph cannot be read at all. (C) Brand B sold nothing.
(D) Brand A and Brand B sold almost the same amount; the truncated axis exaggerated a tiny 2-unit gap.

Solutions — Mathematics (Class 7), Chapter 13: Connecting the Dots

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	C	31	B	41	B	51	A
2	D	12	B	22	B	32	C	42	D	52	B
3	C	13	D	23	A	33	D	43	B	53	D
4	A	14	A	24	D	34	C	44	A	54	C
5	C	15	C	25	B	35	B	45	C	55	A
6	D	16	A	26	C	36	A	46	B	56	B
7	A	17	B	27	C	37	D	47	D	57	C
8	C	18	D	28	A	38	B	48	A	58	C
9	B	19	C	29	A	39	A	49	B	59	B
10	C	20	A	30	D	40	D	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) A statistical statement reasons from data to a *likely* (not certain) conclusion. "A person 6 ft tall is more likely to be an adult than a toddler" is a likelihood claim from a pattern. The others are definitions/arithmetic facts — certain, not statistical.

2. (D) Shubman played 5 matches, Yashasvi 4. More innings give more chances to pile up runs, so the raw total is biased toward whoever batted more often. A fair comparison divides by the number of matches (the mean). The "bigger total = better" reflex is exactly the trap.

3. (C) Mean = $(0 + 17 + 21 + 90) \div 4 = 128 \div 4 = 32$. (128 is the total, not the mean — option D is the classic forgot-to-divide slip.)

4. (A) Sort: 0, 17, 21, 90. Even count (4), so the median is the average of the two middle values: $(17 + 21) \div 2 = 19$. (32 is the mean; don't confuse the two.)

5. (C) Count frequencies: 5 appears 3 times, 3 appears twice, 7 and 8 once each. The mode is 5.

- 6. (D)** Range = largest – smallest = $90 - 33 = 57$. (90 is just the maximum, a common mistake.)
- 7. (A)** Total of five = mean $\times$ count = $20 \times 5 = 100$. Sum of the four known = $18 + 22 + 15 + 25 = 80$. Fifth = $100 - 80 = 20$.
- 8. (C)** "Which size sells most often" asks for the most frequent value — the mode. Mean or median could land on a half-size nobody wears.
- 9. (B)** The median is the middle of the *sorted* list, so you must sort first. You never need to add or divide for a median.
- 10. (C)** Sort: 3, 4, 5, 7, 9, 11. Even count, two middle values 5 and 7, median = $(5 + 7) \div 2 = 6$.
- 11. (A)** 1 cm = 10 runs, so 4.5 cm = $4.5 \times 10 = 45$ runs.
- 12. (B)** 75 runs $\div$ 10 runs-per-cm = 7.5 cm.
- 13. (D)** Mean = $(0 + 1 + 2 + 3 + 4) \div 5 = 10 \div 5 = 2$. (10 is the total.)
- 14. (A)** Sum = $12 + 15 + 18 + 11 + 14 = 70$; mean = $70 \div 5 = 14$.
- 15. (C)** Frequencies: 4 appears 3 times (most), 7 twice, others once. Mode = 4.
- 16. (A)** Both means are 30, so consistency is judged by spread. A's range = $32 - 28 = 4$; B's range = $58 - 2 = 56$. Smaller range = more consistent, so Player A. Equal means do NOT mean equal consistency.
- 17. (B)** Frequencies: 30 appears 3 times — more than any other. Mode = 30.
- 18. (D)** Sort: 30, 30, 30, 31, 32, 33, 35. Seven values, the 4th is the middle = 31. (The mode 30 is a tempting wrong pick.)
- 19. (C)** Data 60, 60, 70, 80, 80: mean = $350 \div 5 = 70$, median (middle of sorted) = 70 — both true, so C. There are two modes (60 and 80), so A is false; "mode is 70" is false (70 occurs once); range = $80 - 60 = 20$, not 80.
- 20. (A)** New total = old total + 45 = $(4 \times 35) + 45 = 140 + 45 = 185$. New mean = $185 \div 5 = 37$.
- 21. (C)** Adding the same constant to every value raises the mean by that constant: $12 + 4 = 16$. (Multiplying the count is a misread.)
- 22. (B)** Total = $21 \times 8 = 168$. Remove 14: $168 - 14 = 154$. New mean = $154 \div 7 = 22$.

- 23. (A)** Tallest bar = Banana (40 kg). Shortest = Mango (15 kg). Difference = $40 - 15 = 25$ kg.
- 24. (D)** Sort: 7, 9, 12, 20. Even count, median = $(9 + 12) \div 2 = 10.5$.
- 25. (B)** For 5, 5, 5, 5: mean = 5, median = 5, mode = 5 — all equal. Check the others: 3, 4, 5, 7 has mean 4.75; 1, 5, 9 has mean 5 and median 5 but no mode; 4, 5, 6, 8 has mean 5.75 and no mode. Only the all-5 set has mean, median AND mode all equal to 5.
- 26. (C)** The eye judges a bar's value by its height as if it grew from zero. With the axis truncated at 90, the visible part of the 96-bar is far longer than that of the 92-bar, so a 4-unit change looks huge. The real change is only about 4%.
- 27. (C)** The mean uses every value, so the outlier 100 drags it far above the cluster around 4. The mode (4) and median (4) ignore the extreme. So the mean is most distorted.
- 28. (A)** Sum = $8 + 6 + 7 + 9 + 6 + 10 + 5 = 51$. Mean = $51 \div 7 = 7.285... \approx 7.3$.
- 29. (A)** Total = $15 \times 6 = 90$. Known sum = $10 + 12 + 18 + 20 + 14 = 74$. Sixth = $90 - 74 = 16$.
- 30. (D)** 1 symbol = 5 books, $6\frac{1}{2}$ symbols = $6.5 \times 5 = 32.5$ books.
- 31. (B)** Runs 30, 45, 45, 60, 20: 45 occurs twice, every other once. Mode = 45.
- 32. (C)** Data 2, 2, 2, 5, 8, 100: mode = 2; median = $(2 + 5) \div 2 = 3.5$; mean = $119 \div 6 \approx 19.8$. So mode < median < mean — the outlier pulls the mean far right.
- 33. (D)** Cricket gap = $18 - 12 = 6$; Football gap = $|10 - 22| = 12$. Football's gap (12) is larger. (32 is the sum of the two football bars, a distractor.)
- 34. (C)** A value was understated by $56 - 36 = 20$, so the recorded total was 20 too low. Old total = $25 \times 10 = 250$; true total = 270; corrected mean = $270 \div 10 = 27$.
- 35. (B)** Combine totals: $(30 \times 140) + (20 \times 150) = 4200 + 3000 = 7200$; over 50 students = 144 cm. (Averaging 140 and 150 to 145 ignores the unequal group sizes — the trap.)
- 36. (A)** Six sorted values; median = average of the 3rd and 4th = $(x + 13) \div 2 = 11$, so $x + 13 = 22$, $x = 9$.
- 37. (D)** Each of 2, 3 and 5 appears exactly twice; no single value is most frequent, so there is no single mode.
- 38. (B)** Mean = $(80 + 120 + 100 + 60) \div 4 = 360 \div 4 = 90$ mm.
- 39. (A)** Data 5, 8, 8, 11, 13: mean = $45 \div 5 = 9$; median (3rd value) = 8; mean – median = $9 - 8 = 1$.

40. (D) First batter's range = $90 - 0 = 90$; second batter's range = $67 - 18 = 49$. Smaller range = more consistent, so the second batter (range 49). Option B attaches "range 49" to the wrong batter — a swap trap.

41. (B) Combined total = $(9 \times 30) + (3 \times 42) = 270 + 126 = 396$; over 12 = 33.

42. (D) Changing the 14 to 27 re-sorts to 10, 16, 16, 19, 21, 27. The mode stays 16 (still the only repeated value). The mean, range (was 11, now 17) and median (was 16, now $(16+19)/2 = 17.5$) all change. So the mode is unchanged.

43. (B) Ages 11, 12, 12, 13, 13, 13, 14, 16: 13 appears most $\rightarrow$ mode = 13. Eight values, median = average of 4th and 5th = $(13 + 13) \div 2 = 13$. So mode = 13, median = 13.

44. (A) Pets 0, 1, 0, 2, 1, 0, 3, 1, 0: 0 appears 4 times, 1 appears 3 times. Mode = 0. (Picking 1 because "1 pet" feels typical is the slip.)

45. (C) Five consecutive whole numbers with mean 12 are centred on 12: 10, 11, 12, 13, 14. The largest is 14. (For an odd run of consecutive numbers, the mean equals the middle value.)

46. (B) Mean daily saving = $(20 + 0 + 40 + 30 + 10) \div 5 = 100 \div 5 = ₹20$. The zero still counts as one of the 5 days — dividing by 4 is the trap.

47. (D) Sort: 7, 9, 9, 12, 14, 18, 21. Seven values, the 4th is the middle = 12.

48. (A) When the mean sits well above the median, a few large outliers are pulling the mean up while the median (the middle) stays near the cluster. Sorting doesn't change the mean, and equal modes/means aren't implied.

49. (B) Total = $8 \times 5 = 40$. Known sum = $6 + 9 + 7 + 11 = 33$. Fifth = $40 - 33 = 7$.

50. (D) $X = 7 \times 4 = 28$ cars; $Y = 5.5 \times 4 = 22$ cars. Difference = $28 - 22 = 6$ cars. (Subtracting the symbols, $7 - 5.5 = 1.5$, then forgetting to scale gives the C trap.)

51. (A) Data 3, 7, 7, 7, 9, 11, 12: mode = 7 (three times); median = 4th value = 7; mean = $(3 + 7 + 7 + 7 + 9 + 11 + 12) \div 7 = 56 \div 7 = 8$. So mode 7, median 7, mean 8 $\rightarrow$ "7, 7, 8", which is option A. (Option D, "7, 7, 7", wrongly takes the mean to be 7 too.)

52. (B) Equal means (60) but Section P's range is 8 versus Q's 40, so P's marks are bunched closely — more consistent / less spread. Larger range (Q) means more scatter, not tighter. Equal means never imply identical marks.

53. (D) Total = $13 \times 5 = 65$. Known sum = $6 + 10 + 14 + 20 = 50$. $x = 65 - 50 = 15$.

54. (C) 1 unit = 2 students, 9 units = $9 \times 2 = 18$ students.

55. (A) Marks 10, 12, 12, 14, 15, 15, 15, 18, 19: mode = 15 (three times); nine values, median = 5th = 15. Mode – median = 15 – 15 = 0.

56. (B) Range = largest – smallest, so largest = smallest + range = 8 + 12 = 20. (The mean 14 is a distractor not needed here.)

57. (C) Total = $(3 \times 10) + (2 \times 15) = 30 + 30 = 60$; over 5 numbers = $60 \div 5 = 12$. (Averaging 10 and 15 to 12.5 ignores the unequal counts — the trap.)

58. (C) Trees 30, 45, 60, 45: mode = 45 (appears twice). Sort 30, 45, 45, 60; median = $(45 + 45) \div 2 = 45$. Both the mode and the median equal 45.

59. (B) Mean 11 over 6 numbers $\rightarrow$ total = 66. Known sum = $5 + 9 + 11 + 11 + 14 = 50$, so $x = 66 - 50 = 16$. The mode stays 11 (11 still appears twice; 16 once), consistent with the conditions.

60. (D) With the axis starting at 50, only the slivers above 50 are shown: 54 $\rightarrow$ 4 units, 52 $\rightarrow$ 2 units, so one bar looks twice the other even though the real values (54 vs 52) are almost equal. The honest reading: nearly the same sales; the truncated axis exaggerated a 2-unit gap.

End of solutions.